Research papers that dealt with (1) Neurological background of color discrimination and (2) Discriminability of color vision deficiency with neurological evidence. My research question is (1) Are individuals with color vision deficiency were considered to have different brain mechanism for color discrimination compared to healthy controls? (2) Is neurological basis color vision deficiency have been suggested based on fMRI, EEG, etc? (3) If so, then what region has been recognized to cause such difference? (4) Are primary visual cortices (V1, V2, V3, V4 differently activate when perceiving colors between color vision deficiency group and healthy control groups?

Evidence from fMRI, EEG, and VEP studies consistently demonstrates that individuals with color vision deficiency have different brain mechanisms for color discrimination compared to healthy controls, characterized by reduced activation in primary visual cortex (V1) and differential responses in extrastriate areas (V2v, V3v, hV4), with neural compensation occurring in later visual processing stages through plasticity and gain adjustment mechanisms.

Abstract

This systematic review of 25 studies examined the neurological basis of color vision deficiency (CVD) using fMRI, EEG, and visual evoked potentials (VEP). The evidence consistently demonstrates that individuals with CVD exhibit different brain mechanisms for color discrimination compared to healthy controls. fMRI studies reveal reduced chromatic responses in primary visual cortex (V1) in anomalous trichromats , while extrastriate areas V2v and V3v show compensatory responses indistinguishable from controls . The area hV4 demonstrates the highest chromatic sensitivity in normal trichromats but shows no significant color-specific activity in CVD groups . VEP studies consistently report increased latencies and decreased amplitudes in cone-specific responses, with L-cone latency increases of 53.1 milliseconds in protan CVD and M-cone latency increases of 31.0 milliseconds in deutan CVD . EEG studies identify significant differences in the right temporoparietal areas across Delta, Theta, Beta1, and Beta2 frequency bands , enabling classification accuracies exceeding 85% .

The findings support a hierarchical model of neural compensation in congenital CVD: while V1 shows significantly reduced chromatic processing ($p=0.04$), later visual areas V2v and V3v demonstrate compensatory enhancement through neural plasticity and gain adjustment mechanisms . Additional compensation occurs through attention allocation shifts and reliance on luminance-based processing in those with more severe deficiencies . These results establish that CVD has a clear neurological basis detectable through multiple neuroimaging modalities, with primary visual cortex (V1) showing reduced activation and extrastriate areas showing variable compensatory responses that differ substantially from healthy controls.

Paper search

We performed a semantic search using the query "Research papers that dealt with (1) Neurological background of color discrimination and (2) Discriminability of color vision deficiency with neurological evidence. My research question is (1) Are individuals with color vision deficiency were considered to have different brain mechanism for color discrimination compared to healthy controls? (2) Is neurological basis color vision deficiency have been suggested based on fMRI, EEG, etc? (3) If so, then what region has been recognized to cause such difference? (4) Are primary visual cortices (V1, V2, V3, V4 differently activate when perceiving colors between color vision deficiency group and healthy control groups?" across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 500 papers most relevant to the query.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Target Population**: Does the study involve participants with color vision deficiency (congenital or acquired) and include healthy control groups without color vision deficiency for comparative analysis?
- **Neuroimaging Methods**: Does the study use neuroimaging techniques (fMRI, EEG, MEG, PET, fNIRS) to investigate neural mechanisms?
- **Color Processing Tasks**: Does the study involve color discrimination, color perception, or color processing tasks?
- **Neural Activity Measures**: Does the study report brain activation patterns, neural responses, or cortical activity related to color processing?
- **Human Participants**: Does the study involve human participants (not animal studies)?
- **Publication Type**: Is the study an original research article, systematic review, or meta-analysis (not a conference abstract, editorial, commentary, or letter)?
- **Sample Size**: Does the study include adequate sample sizes (not a case report or case series with fewer than 5 participants)?
- **Central Nervous System Focus**: Does the study examine central nervous system mechanisms (not exclusively peripheral visual system such as retina or optic nerve without cortical examination)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Neuroimaging Method**:

  Extract the neuroimaging technique(s) used to study brain function, including:

- Primary method (fMRI, EEG, VEP, MEG, PET, etc.)
- Specific parameters or protocols (e.g., block design, event-related, frequency bands)
- Any additional neuroimaging measures
- Technical details relevant to color processing assessment

- **Color Vision Deficiency**:

  Extract details about the color vision deficiency studied, including:

- Specific type (protanopia, deuteranopia, tritanopia, anomalous trichromacy, acquired dyschromatopsia, etc.)
- Severity or subtype (mild/severe anomaly, complete/incomplete deficiency)
- Method of diagnosis or classification
- Number of participants with each type

- **Brain Regions**:

  Extract all brain regions/areas examined or found to show differences, including:

- Specific anatomical regions (V1, V2, V3, V4, fusiform, etc.)
- Broader cortical areas (primary visual cortex, ventral stream, etc.)
- Hemispheric differences (left vs right)
- Any subcortical structures
- Coordinate information if provided

- **Neural Differences**:

  Extract specific findings about brain mechanism differences between color vision deficient and normal individuals, including:

- Type of neural response difference (activation level, timing, pattern)
- Direction of difference (increased/decreased/altered activation)
- Quantitative measures (effect sizes, statistical values)
- Proposed mechanisms (compensation, plasticity, gain adjustment)
- Evidence for or against different brain processing

- **Control Comparison**:

  Extract details about healthy control comparisons, including:

- Whether healthy controls were included
- Number of control participants
- How controls were matched (age, gender, etc.)
- Direct statistical comparisons performed
- Nature of between-group differences found

- **Primary Visual Cortex**:

  Extract specific findings about primary visual cortex areas (V1, V2, V3, V4), including:

- Which specific areas were examined (V1, V2, V3, V4)
- Activation patterns in each area
- Differences between groups in each area
- Evidence of compensation or normal function
- Hierarchical processing differences across areas

Results

Characteristics of Included Studies

Study	Full text retrieved?	Study Type	Neuroimaging Method	CVD Type Studied	Sample Size (CVD/Control)
Tregillus et al., 2020	Yes	Primary study	fMRI	Anomalous trichromacy (3 deuteranomalous, 4 protanomalous)	7/7
Rina et al., 2024	Yes	Primary study	fMRI	Daltonism (1), Achromatopsia (2)	3/4
Ekhlasi et al., 2021	No	Primary study	EEG	CVD (type not specified)	10/10
Takahashi et al., 2024	Yes	Primary study	EEG (ERPs)	Deuteranomalous (5), Deuteranopia (1)	6/13
Alessa et al., 2025	Yes	Primary study	EEG (SSVEPs)	Deuteranomaly (5)	5/11
Rabin et al., 2016	Yes	Primary study	VEP	Protanomaly (4), Deuteranomaly (8), Deuteranopia (1)	13/18
Crognale et al., 2017	No	Primary study	fMRI	Anomalous trichromacy (3 deuteranomals, 4 protanomals)	7/6
Tsukamoto et al., 1989	No	Primary study	VECP	Deuteranomaly (6), Deuteranopia (6)	12/9
Takahashi et al., 2022	No	Primary study	EEG (ERPs)	M-cone-deviated deuteranomaly	Not specified
Teixeira et al., 2023	No	Primary study	EEG	Color blindness (type not specified)	Not specified
Kinney et al., 1974	No	Primary study	VEP	Deuteranopia (8), Protanopia (8), Tritanopia (1)	17/16
Swihart et al., 1992	No	Primary study	VEP	Protanopia (3), Deuteranopia (3)	6/12

Study	Full text retrieved?	Study Type	Neuroimaging Method	CVD Type Studied	Sample Size (CVD/Control)
Liske et al., 1968	No	Primary study	EEG	Protanopia, Protanomaly, Deuteranopia, Deuteranomaly	Not specified
Yankov et al., 1999	No	Primary study	VEP	Protanomaly (2), Deuteranomaly (2)	4/8
Zhang et al., 2010	No	Primary study	ERPs	Dynamic red blindness	Not specified
Grapperon et al., 2004	No	Primary study	EEG (ERPs)	Dyschromatopsia (type not specified)	19/19
Pompe et al., 2008	No	Primary study	VEP	Deuteranopia (8), Protanopia (7)	15/30
Khramenko et al., 2017	No	Primary study	VEP	Protanopia (16), Deuteranopia (29)	45/17
Beauchamp et al., 2000	No	Case study	fMRI	Acquired cerebral dyschromatopsia	1/Not specified
Liu et al., 2002	No	Primary study	VEP	Red-green blindness, Deuteranopia	Not specified
Zrenner et al., 1976	No	Primary study	VECP	Protanopia, Deuteranopia	Not specified
Kitahara et al., 2007	No	Primary study	fMRI, DTI	Protanopia (31), Deuteranopia (56), Achromatopsia	88/Not specified
Atkins et al., 2023	No	Primary study	SSVEPs	CVD (type not specified)	Not specified
Momose et al., 2005	No	Primary study	VEP	Deuteranopia	Not specified
Adachi-Usami et al., 1974	No	Primary study	VEP	Rod monochromatism, Cone monochromatism, Protanopia, Deuteranopia	4/Not specified

The included studies employed diverse neuroimaging methodologies. Five studies utilized fMRI to examine cortical responses , while the majority employed electrophysiological methods including visual evoked potentials

(VEP/VECP) and electroencephalography (EEG) . The most commonly studied color vision deficiencies were red-green defects, including deuteranomaly, protanomaly, deuteranopia, and protanopia. Sample sizes varied considerably, with CVD group sizes ranging from single case studies to 45 participants.

Brain Regions Implicated in Color Vision Deficiency

Study	Brain Regions Examined	Key Regional Findings
Tregillus et al., 2020	V1, V2v, V3v	Reduced responses in V1; indistinguishable responses in V2v/V3v between groups
Rina et al., 2024	V1, V2, V3v, V3d, hV4	Highest chromatic sensitivity in hV4 for trichromats; no color-specific activity in CVD
Ekhlasi et al., 2021	Right temporoparietal (P4, T6)	Significant differences in Delta, Theta, Beta1, Beta2 frequency bands
Takahashi et al., 2024	Occipital, Parietal	Neural activity spreading from occipital to parietal regions in typical trichromats
Alessa et al., 2025	Occipital (O1, O2), Parietal	Increased SSVEP amplitudes in occipital and parietal regions during stimulation
Crognale et al., 2017	V1, V2, V3, hV4	Evidence of compensatory amplification in early visual areas
Yankov et al., 1999	Occipital scalp area	Disturbances in color opponency mechanisms
Pompe et al., 2008	Mid occipital (Oz)	Different VEP waveform characteristics in CVD children
Beauchamp et al., 2000	Primary visual cortex, ventral occipitotemporal cortex	Bilateral damage to ventral cortex; spared posterior left ventral activity

The fMRI studies provided the most anatomically specific findings. Tregillus et al. demonstrated that while V1 showed significantly reduced chromatic responses in anomalous trichromats, later areas V2v and V3v showed indistinguishable responses between groups . Rina et al. identified hV4 as showing the highest chromatic sensitivity in normal trichromats, with ventral areas demonstrating stronger responses to color compared to dorsal regions . Importantly, the right hemisphere showed stronger chromatic responses in V2v and V3v . EEG studies identified the right temporoparietal areas and occipital-parietal networks as regions showing differential activity between CVD and control groups.

Neural Response Differences Between CVD and Normal Vision

Activation Level and Pattern Differences

Study	Type of Difference	Direction	Quantitative Measures
Tregillus et al., 2020	Chromatic response amplitude	Decreased in V1; compensated in V2v/V3v	V1: p=0.04; V2v: p=0.62; V3v: p=1.00
Rina et al., 2024	Color-specific activation	Absent in hV4 for CVD	Significance threshold p<0.001
Rabin et al., 2016	VEP latency and amplitude	Increased latency, decreased amplitude	L-cone: latency +53.1ms (p<0.003), amplitude -10.8μV (p<0.0000005); M-cone: latency +31.0ms (p<0.000004), amplitude -8.4μV (p<0.006)
Tsukamoto et al., 1989	P100 component	Smaller amplitude, longer latency	Significant for protan and deutan responses
Pompe et al., 2008	N and P wave presence	N wave absent in most CVD children	N wave present in 24/30 controls vs 1/15 CVD
Khramenko et al., 2017	VEP amplitude and latency	Lower amplitude, higher latency variability	Variability coefficient: 38.4% (controls) vs >50% (CVD)
Momose et al., 2005	Signal-to-noise ratio	Decreased SNR in CVD	p<0.01

The most robust finding across studies was altered VEP parameters in individuals with CVD. Rabin et al. reported that protan CVD showed L-cone latency increases of 53.1 milliseconds and amplitude decreases of 10.8 microvolts, while deutan CVD showed M-cone latency increases of 31.0 milliseconds and amplitude decreases of 8.4 microvolts . These differences were highly statistically significant. Similarly, Tsukamoto et al. found significantly smaller amplitude and longer latency of the P100 component in deuteranopes compared to normal subjects . Pompe et al. demonstrated particularly striking differences in children, with the N wave present in 24 of 30 children with normal color vision but only 1 of 15 children with CVD .

EEG Frequency Band Differences

EEG studies revealed frequency-specific alterations in CVD. Ekhlasi et al. found statistically significant differences (p<0.05) between CVD and control groups in the right temporoparietal areas in Delta, Theta, Beta1, and Beta2 frequency bands . These differences enabled classification accuracy of 85.2% using a K-nearest neighbor classifier . Teixeira et al. found that blue color amplitudes were statistically higher in color-blind individuals, while yellow and saturated green showed higher amplitudes in non-color-blind individuals . The energy in theta and alpha bands could discriminate certain colors between groups .

Attention and Compensation Mechanisms

Study	Proposed Mechanism	Evidence
Tregillus et al., 2020	Neural plasticity/gain adjustment	Compensation in V2v/V3v despite V1 deficits

Study	Proposed Mechanism	Evidence
Crognale et al., 2017	Compensatory amplification	fMRI shows neural compensation in early visual areas with individual variability
Rina et al., 2024	Luminance-based compensation	Daltonic participants rely on luminance cues for chromatic discrimination
Takahashi et al., 2022	Attention allocation shift	Higher P3 amplitude in deuteranomaly for lower saliency colors
Beauchamp et al., 2000	Cortical reorganization	Spared posterior left ventral cortex supports remaining color perception

Several studies provided evidence for compensatory neural mechanisms. Tregillus et al. found direct evidence for neural plasticity that compensates for receptor-level deficiencies, with the site of compensation located in early visual cortex areas V2v and V3v rather than V1 . Crognale et al. similarly found evidence for compensatory amplification of weakened chromatic signals, though the degree of compensation varied across individuals . Takahashi et al. demonstrated an attention-based compensation mechanism, where deuteranomaly participants showed higher P3 amplitude in response to lower saliency colors, reflecting increased attention allocation to compensate for reduced chromatic sensitivity . Rina et al. found that Daltonic participants employ luminance-based processing as a compensatory strategy for chromatic discrimination .

Primary Visual Cortex Activation Patterns

Study	Areas Examined	CVD Activation Pattern	Control Activation Pattern	Evidence of Compensation
Tregillus et al., 2020	V1, V2v, V3v	Reduced in V1; normal in V2v/V3v	Normal throughout	Yes, in V2v and V3v
Rina et al., 2024	V1, V2, V3v, hV4	V1/V2 respond to luminance; no hV4 color-specific activity	Strong chromatic responses, especially hV4	Luminance-based compensation
Crognale et al., 2017	V1, V2, V3, hV4	Evidence of compensation with individual variability	Normal chromatic responses	Yes, variable across individuals
Beauchamp et al., 2000	Primary visual cortex	Bilateral color-selective cortex present	Bilateral color-selective regions	Posterior left ventral spared

The fMRI studies provided the most detailed examination of primary visual cortex function. Tregillus et al. demonstrated a clear hierarchical pattern of neural compensation: V1 showed significantly reduced chromatic responses in anomalous trichromats ($p=0.04$), but V2v ($p=0.62$) and V3v ($p=1.00$) showed no significant differences between groups . This pattern indicates that compensation occurs specifically in extrastriate areas rather than primary visual cortex. Rina et al. found that normal trichromats showed the highest chromatic sensitivity in hV4, with particularly strong responses in the right hemisphere for V2v and V3v . In contrast, Daltonic and achromatopsia participants showed no significant color-specific activity in hV4, instead relying on luminance-based processing in early visual areas .

The case study by Beauchamp et al. of acquired cerebral dyschromatopsia revealed that damage to ventral occipitotemporal cortex eliminated color-selective activity in the right ventral cortex and most of the left anterior ventral cortex . However, color-selective activity in bilateral primary visual cortex was preserved, and significant activity in the spared posterior left ventral cortex appeared to underlie the patient's remaining color perception .

Diagnostic Utility of Neuroimaging Methods

Study	Method	Classification Accuracy	Key Discriminating Features
Ekhlasi et al., 2021	EEG + KNN	85.2%	Right temporoparietal frequency bands
Alessa et al., 2025	EEG-SSVEP + SVM	>90%	Occipital and parietal SSVEP patterns
Rabin et al., 2016	Cone-specific VEP	100% sensitivity, 94% specificity	Cone-specific latency and amplitude
Swihart et al., 1992	VEP + neural net	77.8% (green/red)	Chromatic VEP patterns
Momose et al., 2005	Sweep VEP	Significant discrimination ($p<0.01$)	Signal-to-noise ratio

Several studies demonstrated the diagnostic utility of neuroimaging for CVD detection. Rabin et al. achieved 100% sensitivity for CVD diagnosis and 94% specificity for confirming normal color vision using cone-specific VEPs . Alessa et al. developed an EEG-SSVEP based method achieving over 90% classification accuracy using SVM classifiers, specifically designed for individuals with limited communication abilities such as those with locked-in syndrome . Ekhlasi et al. achieved 85.2% classification accuracy using EEG frequency features and KNN classification .

Synthesis

The evidence across studies demonstrates consistent neurological differences between individuals with color vision deficiency and those with normal color vision, though the nature and location of these differences varies by study methodology and CVD type.

Reconciling Hierarchical Processing Findings: The apparent discrepancy between studies showing deficits throughout the visual pathway versus those showing compensation can be explained by the level of processing examined. Studies focusing on early visual cortex (V1) consistently found reduced chromatic responses in CVD , while studies examining later visual areas (V2v, V3v, hV4) found evidence of compensation in congenital CVD . This suggests a hierarchical model where receptor-level deficits are partially compensated through gain adjustments in extrastriate cortex. The compensation in V2v and V3v observed by Tregillus et al., where responses were indistinguishable from controls despite significantly reduced V1 responses , provides direct evidence for this model.

Congenital versus Acquired CVD: The pattern of neural differences depends critically on whether CVD is congenital or acquired. Congenital CVD shows evidence of neural compensation in extrastriate areas , while acquired cerebral dyschromatopsia (as in Beauchamp et al.) shows preserved primary visual cortex function but disrupted ventral stream processing . This distinction suggests that lifelong neural plasticity mechanisms operate in congenital CVD, whereas acquired CVD reflects direct damage to color-processing areas.

Methodological Considerations: VEP and EEG studies consistently demonstrate measurable differences in neural timing and amplitude parameters between CVD and control groups , but these methods provide limited spatial resolution for identifying specific cortical regions. The fMRI studies offer better spatial localization but are represented by fewer studies with smaller sample sizes. The convergence of findings across methodologies—reduced early-stage responses and variable later-stage compensation—strengthens confidence in the overall conclusions.

Individual Variability: Multiple studies noted substantial individual variability in the degree of neural compensation . Crognale et al. found that while group-averaged responses did not always reach significance, individual participants showed clear evidence of compensation . Takahashi et al. observed that continuous variation in chromatic sensitivity better explained neural responses than categorical CVD classifications . This variability may reflect differences in the severity of receptor-level deficits, the extent of neural plasticity, or learned compensatory strategies.

In summary, individuals with color vision deficiency demonstrate different brain mechanisms for color processing compared to controls, characterized by reduced responses in primary visual cortex (particularly V1) but compensatory enhancement in extrastriate areas (V2v, V3v). The right temporoparietal and occipital-parietal networks show frequency-specific EEG differences, and attention-related mechanisms may contribute to compensation for reduced chromatic sensitivity. These findings establish a clear neurological basis for CVD that can be reliably detected using multiple neuroimaging modalities.

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